

AI-Based Road Surface Defect Monitoring System

Smart City Road Management

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Presentation Outline

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- 3 Proposed Solution
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Problem Statement

Urban Road Challenges

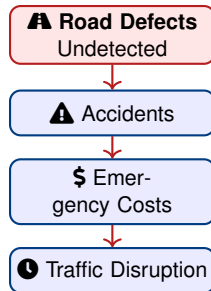
- Municipalities manage **large infrastructure** networks.
- Defects: **potholes, cracks, edge erosion, drainage**.
- Affects vehicle safety, travel quality, & maintenance budget.

Current Inspection Failures

- Manual periodic surveys $\Rightarrow$ **incomplete visibility**.
- Defects go undetected between cycles.
- Leads to accidents, emergency repairs, disruptions.
- No spatially-indexed, severity-based defect database.

Requirement

An **automated, continuous, data-driven** road defect detection and prioritization system for smart city maintenance planning.



Research Gaps & Key Challenges



Imaging Challenges

- **Light & Shadow Variation**
Inconsistent illumination degrades features.
- **Weather Changes**
Rain/fog alter surface appearance.
- **Blurry Images**
Vehicle vibration causes motion blur.



System-Level Gaps

- **Precision vs. Latency**
Models too slow for real-time edge.
- **No Priority System**
No severity-indexed defect triage.
- **No Geo-tagged Database**
Absence of spatially-indexed records.

Gap Summary

Challenge	Prior Work	This Work
Real-time edge inference	×	✓
Weather robustness	Partial	✓
Defect prioritisation	×	✓
Geo-tagged database	×	✓

Proposed Solutions

Dynamic Frame Dropping

- Frame rate adapts to **vehicle speed**.
- Reduces redundant data ingestion.
- Preserves frames at high speeds.
- Reduces overall compute load.

Blur Mitigation Pipeline

- Laplacian variance filters **blurry frames**.
- Temporal frame averaging for static zones.
- Deblur preprocessing for critical images.

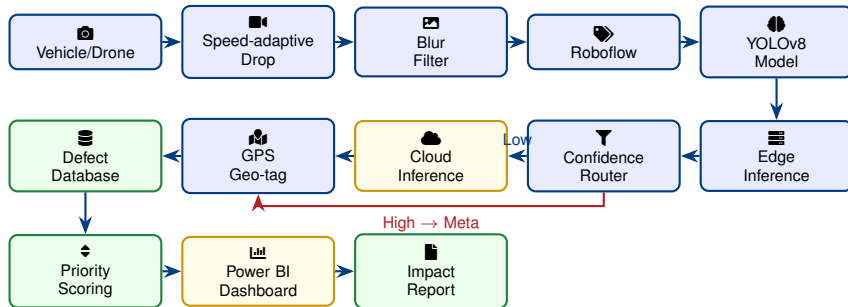
Confidence-Based Routing

- High-confidence: **metadata saved only**.
- Low-confidence: sent to cloud for re-inference.
- Reduces bandwidth by **~60–70%**.

Defect Priority Scoring

- Tiers: **Critical / Moderate**.
- Score = defect class + area + frequency.
- Dashboards with **geo-ranked queues**.

System Architecture Overview



Architecture: Milestone Breakdown

M1 – Data Collection Architecture

- Vehicle & drone camera survey protocols.
- Roboflow annotation & severity tiers.
- Overall inference integration plan.

M2 – Annotation & Model Training

- Annotate images: defect types & severities.
- Train YOLOv8 object detection model.
- Evaluate on representative test samples.

M3 – Batch Inference & Geo-Tagging

- Detect defects from survey images.
- Associate GPS metadata to detections.
- Output: geo-tagged defect database.

M4 – Power BI Dashboard

- Road health scores by zone/corridor.
- Defect severity heatmaps.
- Maintenance priority lists.

M5 – Urban Planning Impact Report

- Demonstrate improved prioritization.
- Estimate reduction in emergency costs vs. planned maintenance.

Key Deliverable Stack

Roboflow → YOLOv8 → Python Pipeline
→ Geo-tagged DB → Power BI

Dataset 1: Vehicle-Mounted Camera (RDD-2022)

- > 47,000 ground-level road images from moving vehicles.
- Collected across Japan, India, Norway, USA, China.
- Captures close-up texture: cracks, potholes, surface wear.
- Trained independently on ground-perspective features.
[\[RDD-2022\]](#)

Dataset 2: Drone Survey (Top-Down)

- High-resolution aerial imagery from UAV surveys.
- Wider field-of-view: road edges, drainage, manholes, large cracks.
- Trained separately on top-down perspective features.
[\[Drone Survey\]](#)

Defect Categories

Code	Type	Source
D00	Severe Cracks	Both
D10	Mild Cracks	Both
D20	Severe Potholes	Both
D40	Mild Potholes	Both
D50	Manholes	Drone only
D60	Smooth roads	Both

Quality Challenges Addressed

- **Illumination:** HSV augmentation.
- **Weather:** Synthetic rain/fog overlays.
- **Imbalance:** Targeted re-sampling per class.

Database Output Schema

Field	Type
Timestamp	ISO 8601
GPS Coords	Lat, Lon
Defect Class	Enum (D00–D60)
Severity Tier	Critical/Mod/Minor
Confidence	Float [0,1]
Bounding Box	Pixel coords
Source Model	Vehicle / Drone
Zone ID	String

Dataset Samples — Vehicle-Mounted Camera

Ground-Level Samples (RCSD) — Captured from Moving Vehicle



Smooth Road



Crack — Mild



Crack — Severe



Pothole — Mild



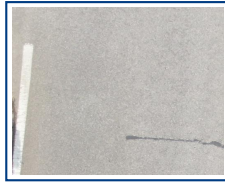
Pothole — Severe

Dataset Samples — Drone Survey

Drone Samples (RDD-2022) — Captured from Drones



Smooth Road



Crack — Mild



Crack — Severe



Pothole — Mild

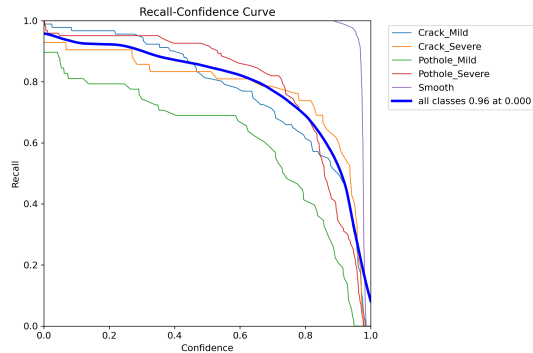
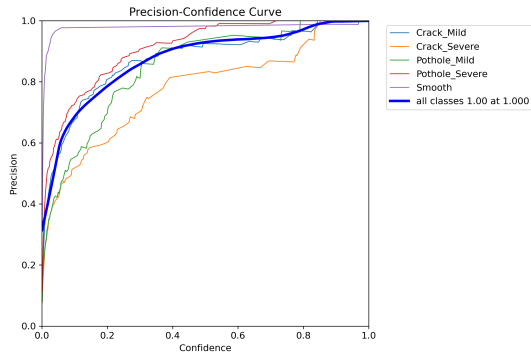


Pothole — Severe

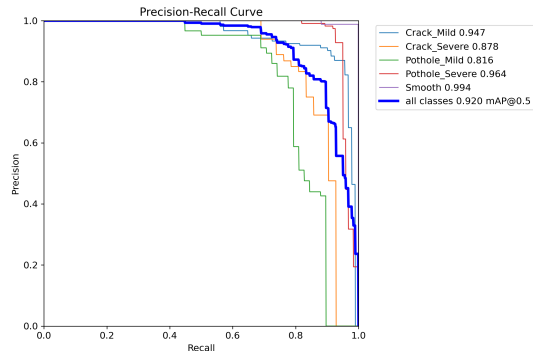
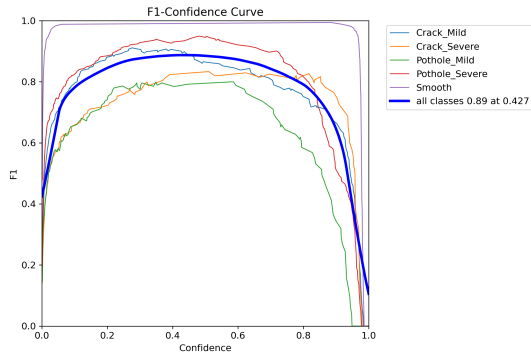


Manhole

Car Model: Precision & Recall Curves

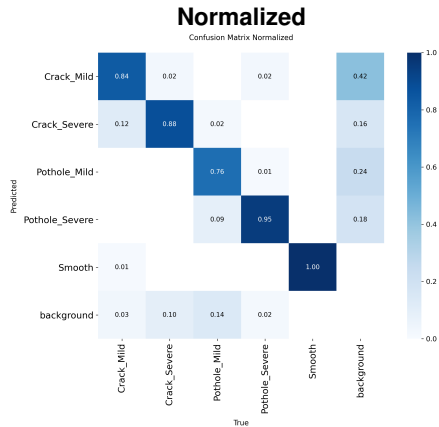
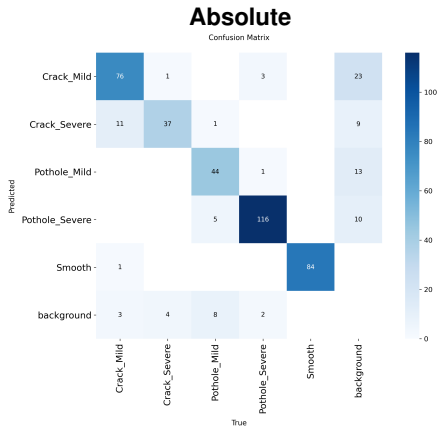


Car Model: F1 & Precision-Recall (PR) Curves

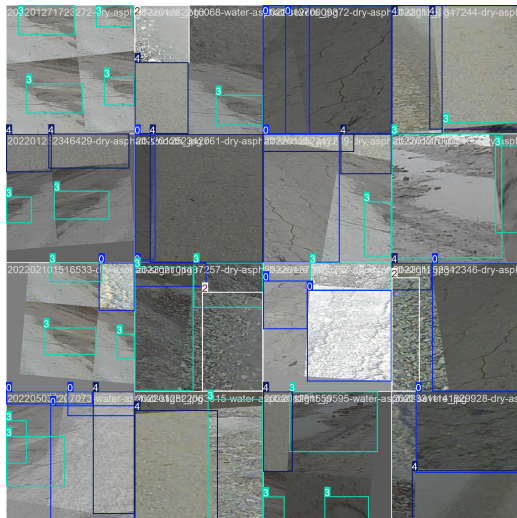


mAP@0.5: 0.920

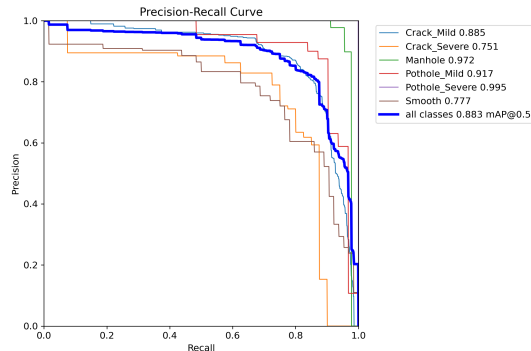
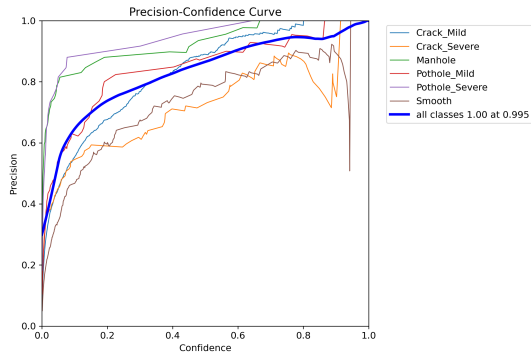
Car Model: Confusion Matrices



Car Model: Training Batch Predictions

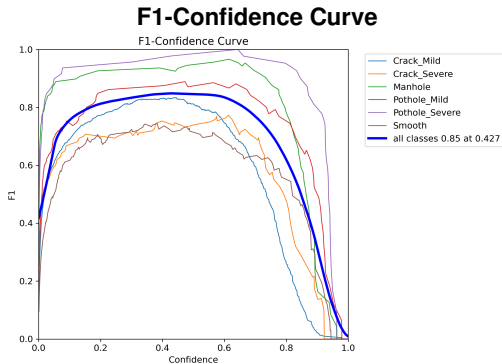


Drone Model: Precision & PR Curves

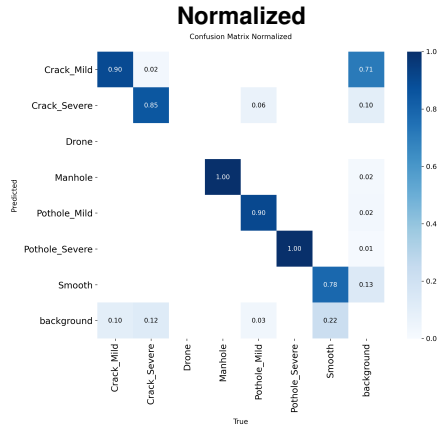
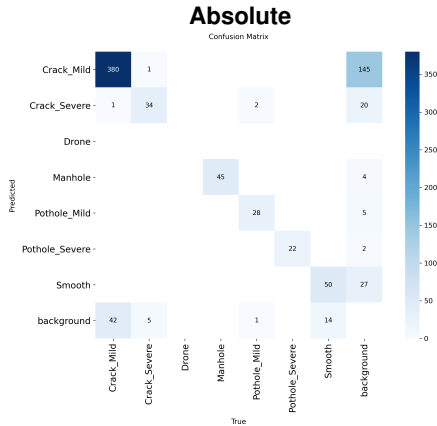


mAP@0.5: 0.883

Drone Model: F1 Curve & Training Batch



Drone Model: Confusion Matrices



Time-Optimized Fleet Routing & Automated Dispatch

Algorithmic Logistics

- **Greedy Nearest-Neighbor:** Utilizes the Haversine formula to compute true distances between active fleets and defects.
- **Dynamic Travel ETA:** Calculates transit times assuming urban traffic constraints ($v \approx 20$ km/h).
- **Dynamic Repair Duration:** Factors variable completion times based on defect severity:
 - ▷ Severe Potholes: +3.5 hours
 - ▷ Severe Cracks: +2.0 hours
 - ▷ Mild Potholes: +1.0 hour

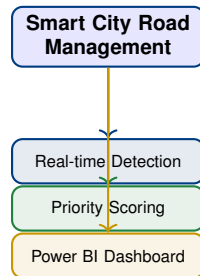
Global Priority Workflow

- 1 **Batch Processing:** Ingests multiple frames simultaneously.
- 2 **Priority Queueing:** Sorts jobs sequentially from *Critical* to *Low*.
- 3 **Time-Optimization:** Assigns the team that will reach the site and finish the job **earliest**, stacking ETAs for busy crews.
- 4 **Auto-Completion:** Monitors system time and transitions job states (*Pending* → *In Progress* → *Completed*) automatically.

Summary & Conclusion

What We Address

- ✓ Blur rejection before model inference
- ✓ Confidence-based bandwidth routing
- ✓ Severity-ranked priority queue
- ✓ Geo-tagged DB + Analytics
- ✓ Global Fleet Dispatch



Thank You

Questions & Discussion

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